

SMART URBAN MOBILITY

& DEMAND INTELLIGENCE DASHBOARD

Milestone 3 - Demand Intelligence, Mobility Analytics & Forecasting

Project	Smart Urban Mobility & Traffic Intelligence
Milestone	Milestone 3
Tool	Microsoft Power BI Desktop
Base dataset	300-record mobility dataset used in Milestones 1 and 2
Prepared by	Tanavi Vadagole

Milestone 3 focus

I extended the same Power BI project from operational mobility monitoring into demand intelligence. The dashboard combines demand, pricing proxies, mobility-mode analysis, weather sensitivity, regional availability and time-based demand analysis in one decision-oriented page.

Documentation prepared for presentation and mentor review

1. Milestone 3 Overview

Milestone 3 builds directly on the star-schema model and interactive dashboard established in the earlier milestones. I retained the same project, dataset and Power BI model and added demand-oriented measures, derived analytical fields and visuals based on the mentor's Milestone 3 reference structure.

Objective	Implementation
Demand intelligence	Measure total passenger demand and study how it varies by date, weather, temperature and mobility mode.
Pricing intelligence	Introduce an estimated fare measure from route distance and an analytical surge multiplier from congestion intensity.
Service/provider analysis	Create a service-provider classification from available vehicle categories and compare demand by provider.
Regional availability	Create a regional classification from directional mobility data and derive a motorcycle-based availability proxy.
Interactive analysis	Use slicers so the same visuals can be explored under different dates, regions and mobility modes.

2. Source Data & Validation

The prepared source workbook contains a 300-record mobility sample. The validation sheet reports 300 output rows representing 5 cities, 60 rows per city, with 0 missing values and 0 duplicate complete rows. The sampling method is documented as a stratified random sample with seed 42.

Data-use principle

I kept the underlying mobility records intact. Fields that were not directly available in the current Power BI semantic model were modeled as derived analytical fields and are explicitly documented as proxies or estimates.

3. Analytical Model & Derived Fields

The central fact table remains FactMobilityTable, connected to Dim_Date, Dim_Location, Dim_Route, Dim_Vehicle and Dim_Weather. This preserves the dimensional structure already used for the earlier milestones and keeps filter propagation predictable.

Derived field	Type	Definition / logic
Estimated Fare	Calculated column	Base fare of 50 plus 12 per kilometre of route distance.
Average Fare	Measure	Average of Estimated Fare under the current filter context.
Surge Multiplier	Calculated column	2.0 when congestion index >=80; 1.5 when >=60; 1.25 when >=40; otherwise 1.0.
Average Surge	Measure	Average of Surge Multiplier under the current filter context.
Service Provider	Calculated column	Maps vehicle categories to analytical provider groups such as Public Transit, Taxi Service and Electric Mobility.
Mobility Mode	Calculated column	Maps vehicle categories to Public Transit, Ride-Hailing, Electric Mobility, Private Car and Two-Wheeler groups.
Bike Availability % Measure	Measure	Motorcycle trip count divided by all mobility records in the current filter context; used as an availability proxy.
Region	Calculated column	Maps Direction values Eastbound/Southbound/Westbound/Northbound to East/South/West/North.

4. Key DAX Measures

Total Demand

Total Demand = SUM(FactMobilityTable[Passenger_Count])

Average Fare

Average Fare = AVERAGE(FactMobilityTable[Estimated Fare])

Average Surge

Average Surge = AVERAGE(FactMobilityTable[Surge Multiplier])

Bike Availability % Measure

Bike Availability % Measure = DIVIDE(CALCULATE(COUNTROWS(FactMobilityTable), Dim_Vehicle[Vehicle_Type] = "Motorcycle"), COUNTROWS(FactMobilityTable), 0)

5. Milestone 3 Dashboard Visual Specification

The final page follows the mentor reference structure while keeping the fields and calculations grounded in this project's available model.

Visual	Visual type	Fields / configuration	Purpose
Total Demand	Card	Total Demand	Executive demand KPI.
Average Fare	Card	Average Fare	Pricing proxy for the current selection.
Average Surge	Card	Average Surge	Demand-pressure pricing proxy.
Demand by Service Provider	Clustered column	Service Provider + Total Demand	Compare demand volume across modeled service-provider groups.
Demand by Mobility Mode	Donut	Mobility Mode + Total Demand	Show demand composition across mobility modes.
Temperature vs Demand	Scatter	Temperature_C + Total Demand	Explore the relationship between temperature and demand.
Demand by Weather	Clustered column	Weather + Total Demand	Compare passenger demand under weather conditions.
Bike Availability % by Region	Clustered column	Region + Bike Availability % Measure	Compare the motorcycle-based availability proxy across four regions.
Demand by Date	Line	Date + Total Demand	Show the time pattern of demand across the reporting period.

6. Dashboard Layout

Area	Elements
Header	Dashboard title: Smart Urban Mobility - Demand & Mobility Intelligence
Top KPI row	Total Demand, Average Fare, Average Surge, Date slicer
Middle row	Demand by Service Provider, Demand by Mobility Mode, Temperature vs Demand
Bottom row	Demand by Weather, Bike Availability % by Region, Demand by Date

Presentation note

The Demand by Date visual is the required historical demand trend. Forecasting was explored during development, but it is treated as an enhancement rather than a substitute for the mentor-specified Demand by Date visual.

7. Interactive Features & User Workflow

Control	Field	What I use it for
Date slicer	Dim_Date[Date]	Restrict demand analysis to a selected reporting period.
Region	FactMobilityTable[Region]	Compare the four derived directional regions.
Mobility Mode	FactMobilityTable[Mobility Mode]	Isolate a specific mobility category.
Cross-filtering	All visuals	Selection in one visual updates connected visuals through the Power BI model.

8. Analytical Workflow

Step	Stage	Interpretation
1	Start with demand KPIs	Read Total Demand, Average Fare and Average Surge.
2	Segment demand	Use the Date and categorical filters to define the scenario.
3	Compare providers and modes	Inspect provider demand and the mode-composition donut.
4	Assess external sensitivity	Use weather and temperature visuals to understand demand variation.
5	Inspect regional availability	Compare the motorcycle-based availability proxy across East, South, West and North.
6	Trace demand over time	Use Demand by Date to identify peaks, dips and changes in demand volume.

9. Quality & Validation Checks

Check	Status / evidence
Model integrity	Fact and dimension structure retained from previous milestones.
Measure validation	Demand, fare, surge and availability measures evaluated under the current filter context.
Visual validation	All six mentor-reference analytical visuals are represented on the Milestone 3 page.
Formatting	Percentage formatting applied to Bike Availability %; KPI titles standardized; chart titles simplified.
Usability	Charts positioned in a consistent 3-column grid with a dedicated KPI/header area.

10. Key Analytical Insights Enabled

Theme	What the dashboard enables
Demand concentration	Total passenger demand can be compared by provider, mode and weather category to reveal how demand is distributed.
Temporal behavior	The date trend provides a direct view of high- and low-demand periods within the selected reporting window.
Environmental sensitivity	Temperature and weather visuals allow demand variation to be examined alongside external conditions.
Regional comparison	The derived regional field enables a consistent four-region comparison for the availability proxy.
Pricing context	Estimated fare and surge multiplier provide a structured way to include the pricing-oriented concepts shown in the mentor dashboard.

11. Limitations & Transparent Assumptions

Several Milestone 3 concepts were not direct source columns in the current Power BI model. I therefore modeled them as analytical proxies and documented the assumptions so they are not presented as measured operational facts.

Area	Limitation / assumption
Fare	Estimated from route distance using a fixed base fare and per-kilometre rate; not a billed fare from a ride-hailing platform.
Surge	Derived from Congestion_Index thresholds; not an observed platform surge multiplier.
Service Provider	Derived from Vehicle_Type categories; not an original provider/company field.
Mobility Mode	Derived from Vehicle_Type categories; not the source Transport_Mode field currently exposed in the Power BI model.
Bike Availability	Proxy based on the share of motorcycle mobility records; not physical real-time bike inventory.
Region	Derived from Direction categories to obtain East, South, West and North groups for the mentor-style comparison.

How I would explain this to a mentor

I would explicitly state that the pricing and availability visuals are modeled analytical proxies added to satisfy the requested demand-intelligence structure, while the underlying mobility data and core demand measures remain unchanged.

12. Final Dashboard Preview

Current Milestone 3 dashboard layout captured from Power BI Desktop.

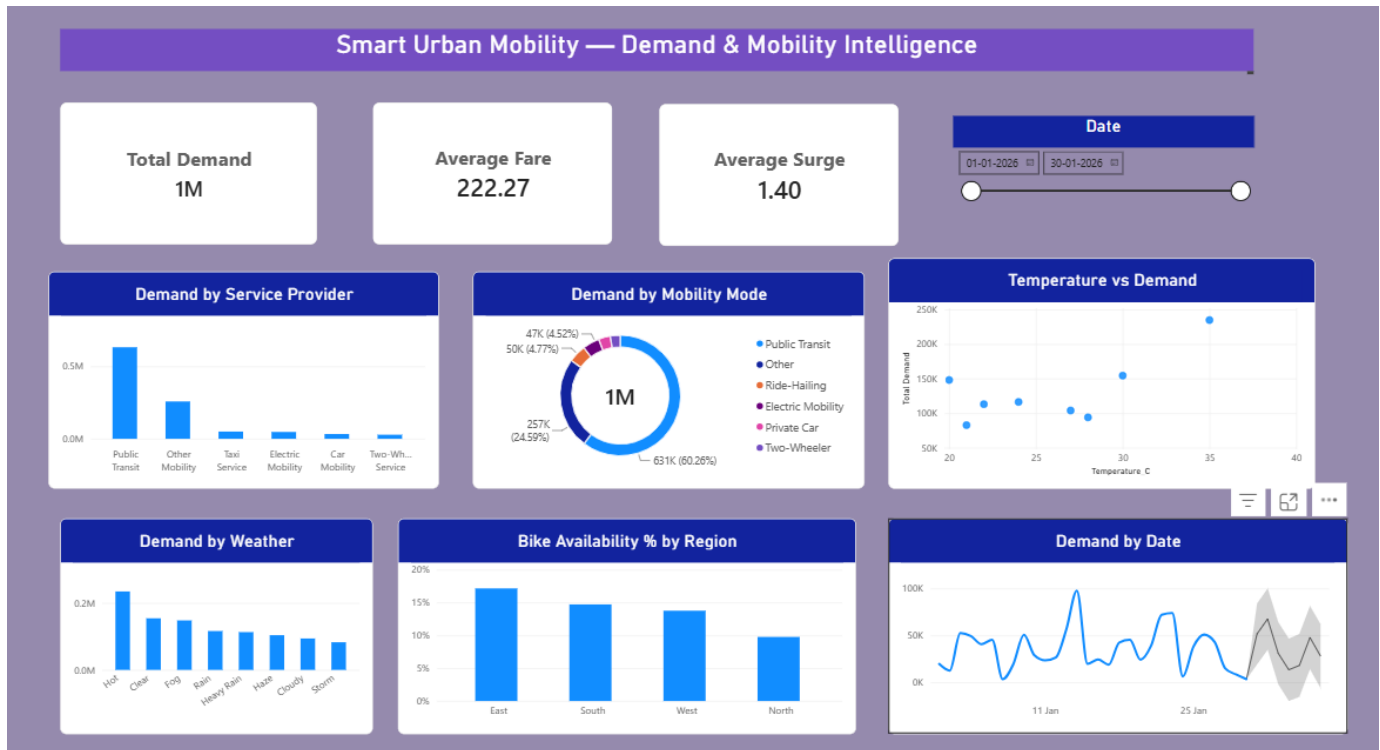


Figure 1. Smart Urban Mobility - Demand & Mobility Intelligence dashboard.

13. Final Presentation Checklist

Requirement	Final status
Three demand KPIs	Completed - Total Demand, Average Fare, Average Surge
Demand by Service Provider	Completed
Demand by Mobility Mode - donut	Completed
Demand by Weather	Completed
Temperature vs Demand	Completed
Bike Availability % by Region	Completed - four derived regions
Demand by Date	Completed - line trend
Professional title and layout	Completed
Transparent documentation of derived fields	Completed
Ready for mentor presentation	Yes

END OF MILESTONE 3 DOCUMENTATION